

24/06/2024

```
1.
#include<stdio.h>
int main()
{
    int num,row,col;
    printf("enter a number");
    scanf("%d",&num);
    for(row=1;row<=num;row++)
    {
        for( col=1;col<=row;col++)
        {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
enter a number 5
*
**
***
****
*****
```

```
2.
#include<stdio.h>
int main()
{
    int num,row,col;
    printf("enter a number");
    scanf("%d",&num);
    for(row=1;row<=num;row++)
    {
        for( col=row;col<=num;col++)
        {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

```
enter a number 5
*****
****
***
**
*
```

```
3.
#include<stdio.h>
int main()
{
    int num,row,col;
    printf("enter a number");
    scanf("%d",&num);
    for(row=1;row<=num;row++)
    {
        for( col=1;col<=num;col++)
        {
            printf("*");
        }
        printf("\n");
    }
}
```

```
    return 0;
}
```

enter a number 5

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*****
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```

4.

```
#include<stdio.h>
int main()
{
    int num,row,col;
    printf("enter a number");
    scanf("%d",&num);
    for(row=1;row<=num;row++)
    {
        printf("*");
    }

    printf("\n");

    return 0;
}
```

enter a number 5

```
*****
```

5.

```
#include<stdio.h>
int main()
{
    int num;
    printf("enter a number: ");
    scanf("%d",&num);
    for(int row=1;row<=num;row++)
    {
        for(int space=1;space<=num-row;space++)
        {
            printf(" ");
        }
        for(int col=1;col<=2*row-1;col++)
        {
            printf("*");
        }
        printf("\n");
    }
}
```

enter a number: 5

```
 *
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*****
*****
```

6.

```
#include<stdio.h>
int main()
{
    int num,row,col;
    printf("enter a number");
    scanf("%d",&num);
    for(row=1;row<=num;row++)
```

```

    {
        printf("*");

        printf("\n");
    }
    return 0;
}

```

enter a number 5

```

*
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```

7.

```

#include<stdio.h>
int main()
{
    int num;
    printf("enter a number: ");
    scanf("%d",&num);
    for(int row=1;row<=num;row++)
    {
        for(int space=1;space<=num-row;space++)
        {
            printf(" ");
        }
        for(int col=1;col<=2*row-1;col++)
        {
            printf("*");
        }
        printf("\n");
    }
    for(int row=num;row>=1;row--)
    {
        for(int space=1;space<=num-row;space++)
        {
            printf(" ");
        }
        for(int col=1;col<=2*row-1;col++)
        {
            printf("*");
        }
        printf("\n");
    }
}

```

enter a number: 5

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*****
***
*

```

8.

```

#include<stdio.h>
int main()
{
    int num;

```

```

printf("enter a number: ");
scanf("%d",&num);
for(int row=1;row<=num;row++)
{
    for(int space=1;space<=num-row;space++)
    {
        printf(" ");
    }
    for(int col=1;col<=2*row;col++)
    {
        if(col==1||col==2*row-1)
        {
            printf("*");
        }
        else
        {
            printf(" ");
        }
    }
    printf("\n");
}
for(int row=num;row>=1;row--)
{
    for(int space=1;space<=num-row;space++)
    {
        printf(" ");
    }
    for(int col=1;col<=2*row;col++)
    {
        if(col==1||col==2*row-1)
        {
            printf("*");
        }
        else
        {
            printf(" ");
        }
    }
    printf("\n");
}
return 0;
}

```

enter a number: 5

```

*
* *
*  *
*    *
*      *
*        *
*          *
*            *
*              *

```

9.

```

#include<stdio.h>
int main()
{
    int num;
    printf("enter a number: ");
    scanf("%d",&num);
    for(int row=1;row<=num;row++)
    {
        for(int col=1;col<=num;col++)
        {

```

```

        if(row==1 || col==1 || row==num || col==num )
        {
            printf("*");
        }
        else
        {
            printf(" ");
        }
    }
    printf("\n");
}
return 0;
}

```

enter a number: 5

```

*****
*   *
*   *
*   *
*****

```

10.

```

#include<stdio.h>
int main()
{
    int num;
    printf("enter a number: ");
    scanf("%d",&num);
    for(int row=1;row<=num;row++)
    {
        for(int col=1;col<=num;col++)
        {
            if( col==num)
            {
                printf("*");
            }
            else
            {
                printf(" ");
            }
        }
        printf("\n");
    }
    return 0;
}

```

enter a number: 5

```

*
*
*
*
*

```

11.

```

#include<stdio.h>
int main()
{
    int num,row,col;
    printf("enter a number");
    scanf("%d",&num);
    for(row=1;row<=num;row++)
    {
        for(int space=1;space<=num-row;space++)

```

```

    {
        printf(" ");
    }
    for( col=1;col<=row;col++)
    {
        printf("*");
    }
    printf("\n");
}
return 0;
}

```

enter a number 5

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*
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****
*****

```

12.

```

#include<stdio.h>
int main()
{
    int num;
    printf("enter a number: ");
    scanf("%d",&num);
    for(int row=1;row<=num;row++)
    {
        for(int space=1;space<=num-row;space++)
        {
            printf(" ");
        }
        for(int col=1;col<=row;col++)
        {
            printf("* ");
        }
        printf("\n");
    }
}

```

enter a number: 5

```

*
* *
* * *
* * * *
* * * * *

```

13.

```

#include<stdio.h>
int main()
{
    int num;
    printf("enter a number: ");
    scanf("%d",&num);
    for(int row=1;row<=num;row++)
    {
        for(int col=1;col<=row;col++)
        {
            if(row==1 || col==1 || row==col || row==num)
            {
                printf("*");
            }
            else

```

```

        {
            printf(" ");
        }

        }
        printf("\n");
    }
    return 0;
}

```

enter a number: 5

```

*
**
* *
* *
*****

```

14.

```

#include<stdio.h>
int main()
{
    int num;
    printf("enter a number: ");
    scanf("%d",&num);
    for(int row=1;row<=num;row++)
    {
        for(int space=1;space<=num-row;space++)
        {
            printf(" ");
        }
        for(int col=1;col<=row;col++)
        {
            if(row==1 || col==1 || row==col || row==num)
            {
                printf("*");
            }
            else
            {
                printf(" ");
            }
        }
        printf("\n");
    }
    return 0;
}

```

enter a number: 5

```

*
**
* *
* *
*****

```

15.

```

#include<stdio.h>
int main()
{
    int num;
    printf("enter a number: ");
    scanf("%d",&num);
    for(int row=1;row<=num;row++)
    {
        for(int space=1;space<=num-row;space++)
        {

```

```

        printf(" ");
    }
    for(int col=1;col<=row;col++)
    {
        printf("%d ",col);
    }
    printf("\n");
}
}

```

enter a number: 5

```

1
1 2
1 2 3
1 2 3 4
1 2 3 4 5

```

16.

```

#include<stdio.h>
int main()
{
    int num;
    printf("enter a number: ");
    scanf("%d",&num);
    for(int row=1;row<=num;row++)
    {
        for(int space=1;space<=num-row;space++)
        {
            printf(" ");
        }
        for(int col=1;col<=2*row;col++)
        {
            if(col==1 || col==2*row-1 || row==num )
            {
                printf("*");
            }
            else
            {
                printf(" ");
            }
        }
        printf("\n");
    }
    return 0;
}

```

enter a number: 5

```

*
* *
*  *
*   *
*****

```

17.

```

#include<stdio.h>
int main()
{
    int num;
    printf("enter a number:");
    scanf("%d",&num);
    for(int r=1;r<=num*2;r+=2)
    {
        for(int c=r;c<=num*2;c++)
        {

```

```

        if(c%2==0)
            printf("%d ",c);
    }
    printf("\n");
}

```

```

enter a number: 5
2 4 6 8 10
4 6 8 10
6 8 10
8 10
10

```

18.

```

#include<stdio.h>
int main()
{
    int num;
    printf("enter a number:");
    scanf("%d",&num);
    for(int r=1;r<num;r++)
    {
        for(int c=r;c<=num;c++)
        {
            printf(" ");
        }
        for(int c=1;c<r;c++)
        {
            printf("*");
        }
        for(int c=1;c<=r;c++)
        {
            printf("*");
        }
        printf("\n");
    }
    for(int r=1;r<=num;r++)
    {
        for(int c=1;c<=r;c++)
        {
            printf(" ");
        }
        for(int c=r;c<num;c++)
        {
            printf("*");
        }
        for(int c=r;c<=num;c++)
        {
            printf("*");
        }
        printf("\n");
    }
}

```

```

enter a number: 5

```

```

*
***
*****
*****
*****
*****
***
*

```

19. diamond.

```
#include<stdio.h>
int main()
{
    int num;
    printf("enter a number: ");
    scanf("%d",&num);
    for(int row=1;row<num;row++)
    {
        for(int space=1;space<=num-row;space++)
        {
            printf(" ");
        }
        for(int col=1;col<=row;col++)
        {
            printf("* ");
        }
        printf("\n");
    }
    for(int row=num;row>=1;row--)
    {
        for(int space=1;space<=num-row;space++)
        {
            printf(" ");
        }
        for(int col=1;col<=row;col++)
        {
            printf("* ");
        }
        printf("\n");
    }
}
```

enter a number: 5

```

*
 * *
* * *
* * * *
* * * * *
* * * *
 * * *
  * *
   *
```

20. identity matrix

```
#include<stdio.h>
#define row 3
#define col 3
int main()
{
    int mat[row][col];
    printf("the identity matrix is: \n");
    for(int i=0;i<row;i++)
    {
        for(int j=0;j<col;j++)
        {
            if(i==j)
            {
                printf("1\t");
            }
            else
            {
                printf("0\t");
            }
        }
    }
}
```

```
    }  
    }  
    printf("\n");  
}  
  
return 0;  
}
```

the identity matrix is:

1	0	0
0	1	0
0	0	1